

Redox Reactions

Question1

100 mL of aqueous solution of 0.05M Cu^{2+} is added to 1 L of 0.1 M KI solution. The resultant solution was titrated with $0.01\text{M Na}_2\text{S}_2\text{O}_3$ solution using starch indicator till blue colour disappeared. What is the volume (in mL) of $\text{Na}_2\text{S}_2\text{O}_3$ used?

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Options:

A.

2000

B.

1000

C.

500

D.

250

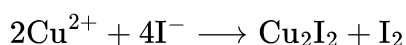
Answer: C

Solution:

Moles of Cu^{2+} : $n_{\text{Cu}^{2+}} = M_{\text{Cu}^{2+}} \times V_{\text{Cu}^{2+}}$

$$= 0.05 \times 0.1 = 0.005 \text{ mol}$$

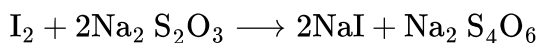
For the reaction,



2 moles of Cu^{2+} produces 1 mol of I_2

$$\text{Moles of } \text{I}_2 : n_{\text{I}_2} = \frac{1}{2} \times n_{\text{Cu}^{2+}} = 0.0025 \text{ mol}$$

For the reaction;



1 mole I_2 react with 2 moles of $\text{Na}_2\text{S}_2\text{O}_3$

$$n_{\text{Na}_2\text{S}_2\text{O}_3} = 2 \times 0.0025 = 0.005 \text{ mol}$$

$$\text{Volume of } \text{Na}_2\text{S}_2\text{O}_3 : V_{\text{Na}_2\text{S}_2\text{O}_3} = \frac{n_{\text{Na}_2\text{S}_2\text{O}_3}}{M_{\text{Na}_2\text{S}_2\text{O}_3}}$$

$$= \frac{0.005}{0.01} = 0.5 \text{ L or } 500 \text{ mL}$$

Question2

H_2O_2 reduces KMnO_4 in acidic medium to ' x ' and in basic medium to ' y '. What are x and y ?

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

$$x = \text{MnO}_2, y = \text{Mn}^{2+}$$

B.

$$x = \text{Mn}^{2+}, y = \text{MnO}_2$$

C.

$$x = \text{MnO}_4^{2-}, y = \text{Mn}^{2+}$$

D.

$$x = \text{MnO}_2, y = \text{MnO}_4^{2-}$$

Answer: B

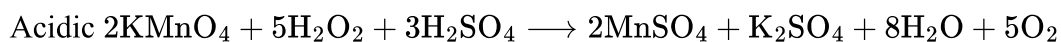
Solution:

In acidic condition, KMnO_4 ($\text{Mn} = +7$ oxidation state) reduced by H_2O_2 to Mn^{2+}

So, $x = \text{Mn}^{2+}$

In basic condition KMnO_4 ($\text{Mn} = +7$ oxidation state)

is reduced by H_2O_2 to MnO_2



Basic



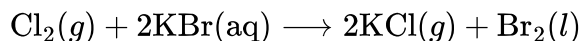
Question3

Which one of the following reactions is not feasible?

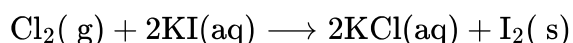
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Options:

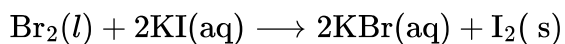
A.



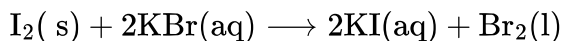
B.



C.



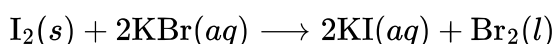
D.



Answer: D

Solution:

Among the given reactions, the reaction



is not feasible because iodine is less reactive than bromine. In a displacement reaction, more reactive element can displace a less reactive element in compound.

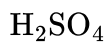
Question4

Which one of the following acts as autocatalyst during titration of KMnO_4 and oxalic acid in presence of dilute H_2SO_4 ?

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Options:

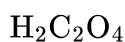
A.



B.



C.



D.



Answer: D

Solution:

MnSO_4 acts as the autocatalyst during the titration of KMnO_4 and oxalic acid.

An autocatalyst is the substance that is formed during a chemical reaction and then speeds up the reaction.

Question5

H_2O_2 with KMnO_4 in acidic medium gives a manganese compound ' X ' and in basic medium gives another manganese compound ' Y '. The oxidation state of manganese in X and Y respectively are

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Options:

A.

+2, +4

B.

+4, +2

C.

+3, +4

D.

+4, +3

Answer: A

Solution:

Oxidation state of Mn in acidic medium

KMnO_4 reduced to Mn^{2+}

Thus, oxidation state = +2

In basic medium,

KMnO_4 reduced to MnO_2

$$0 = x + 2(-2) \Rightarrow x = +4$$

Question6

The volume (in mL) of 10 volume H_2O_2 solution required to completely react with 200 mL of 0.4 M KMnO_4 solution in acidic medium is

Options:

A.

112

B.

336

C.

224

D.

448

Answer: C

Solution:

Normality of KMnO_4 solution,

$$N_{\text{KMnO}_4} = n_{\text{factor}} \times M = 0.4 \times 5 = 2 \text{ N}$$

Normality of H_2O_2

$$N_{\text{H}_2\text{O}_2} = \frac{\text{Volume strength}}{5.6} = \frac{10}{5.6} \text{ N}$$

Using normality equation

$$N_1 V_1 = N_2 V_2$$

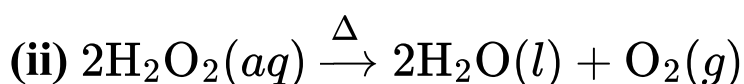
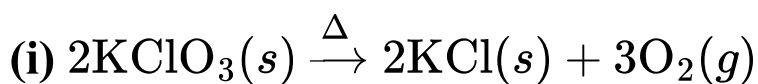
$$N_{\text{KMnO}_4} \times V_{\text{KMnO}_4} = N_{\text{H}_2\text{O}_2} \times V_{\text{H}_2\text{O}_2}$$

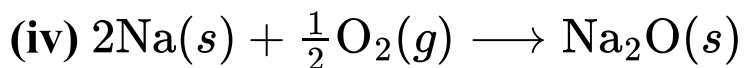
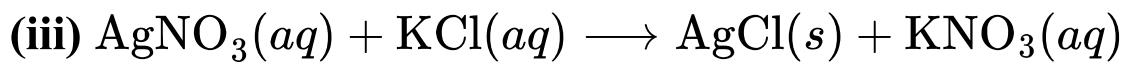
$$2 \times 200 = \frac{10}{5.6} \times V_{\text{H}_2\text{O}_2}$$

$$V_{\text{H}_2\text{O}_2} = 224 \text{ mL}$$

Question7

Observe the following reactions





The number of redox reactions in thsi list is

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Options:

A. 3

B. 4

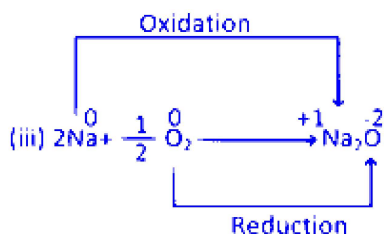
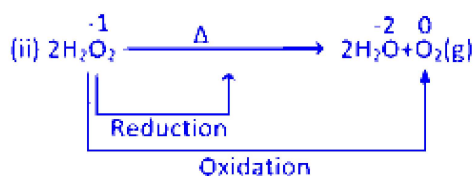
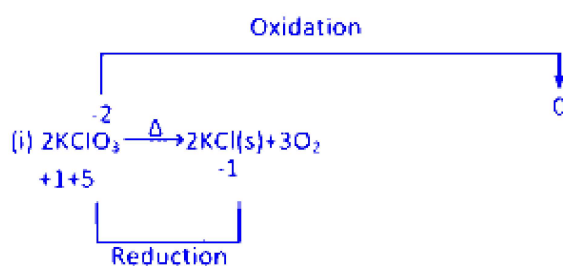
C. 2

D. 1

Answer: A

Solution:

The following reactions are redox reactions.



Reaction (iii) is a double displacement reaction.

Question8

Acidification of chromate gives ' Z '. The oxidation state of chromium in ' Z ' is

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Options:

A. +3

B. +6

C. +7

D. +2

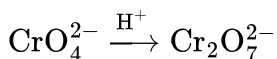
Answer: B

Solution:

The process of acidifying chromate involves the following reaction:



This can be simplified to:



In this reaction, chromate ions (CrO_4^{2-}) are converted into dichromate ions ($\text{Cr}_2\text{O}_7^{2-}$).

In the compound $\text{Na}_2\text{Cr}_2\text{O}_7$, sodium (Na) is in the +1 oxidation state and oxygen (O) is in the -2 oxidation state. To find the oxidation state of chromium (Cr) in $\text{Cr}_2\text{O}_7^{2-}$, we analyze the overall charge balance:

Total charge contributed by two sodium atoms: $2 \times (+1) = +2$.

Each oxygen atom contributes a -2 charge, and there are 7 oxygen atoms: $7 \times (-2) = -14$.

The net charge on the ion $\text{Cr}_2\text{O}_7^{2-}$ is -2.

Setting up the equation for chromium:

$$2 + 2x - 14 = 0$$

Solving for x gives:

$$2x = 12 \Rightarrow x = +6$$

Thus, the oxidation state of chromium in $\text{Na}_2\text{Cr}_2\text{O}_7$ is +6.

Question9

100 mL of 0.1M Fe^{2+} solution was titrated with $\frac{1}{60}$ M $\text{Cr}_2\text{O}_7^{2-}$ solution in acid medium. What is the volume (in L) of $\text{Cr}_2\text{O}_7^{2-}$ solution consumed ?

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Options:

A. 100

B. 10

C. 1

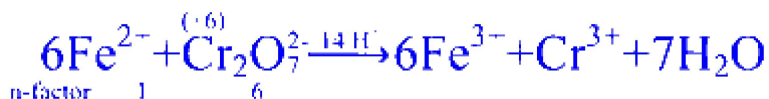
D. 0.1

Answer: D

Solution:

Given, 100 mL of 0.1MFe^{2+} titrated with $\frac{1}{60}\text{MCr}_2\text{O}_7^{2-}$ solution.

The complete reaction is,



$$\Rightarrow \quad 0.1\text{M} \quad \frac{1}{60}\text{M} = 0.1\text{N} = \frac{1}{10}\text{N} \quad [\because N = M \times n\text{-factor}]$$
$$\Rightarrow \quad V\text{ mL} \quad 100\text{ mL} \quad \quad \quad (\text{say})$$

$$\Rightarrow \quad V = \frac{1}{0.1} \times \left(100 \times \frac{1}{10}\right) = 100\text{ mL}$$

So, $V = 0.1\text{ L}$

Question10

Which of the following reactions of KMnO_4 occurs in acidic medium ?

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Options:

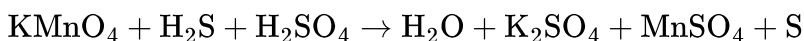
- A. Oxidation of thiosulphate to sulphate
- B. Precipitation of sulphur from H_2S
- C. Oxidation of iodide to iodate
- D. Oxidation of manganous salt to MnO_2

Answer: B

Solution:

When hydrogen sulfide (H_2S) is passed through acidic potassium permanganate (KMnO_4), sulfur is produced.

The balanced chemical equation for this reaction is:



In this reaction, KMnO_4 acts as an oxidizing agent, converting H_2S into sulfur (S).

Question11

Which of the following occurs with KMnO_4 in neutral medium?

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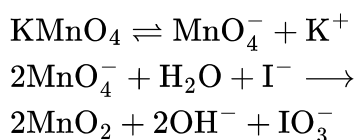
Options:

- A. Oxidation of oxalate ion
- B. Precipitation of sulphur from hydrogen sulphide
- C. Oxidation of Fe^{2+} to Fe^{3+}
- D. Oxidation of iodide to iodate

Answer: D

Solution:

In a neutral medium, the permanganate ion oxidizes iodine to form iodate. Here's the reaction illustrating this process:



This reaction shows how permanganate ions in a neutral solution can facilitate the conversion of iodide ions to iodate ions, with manganese dioxide and hydroxide ions as additional products.

Question12

In the reaction of sulphur with concentrated sulphuric acid, the oxidised product is X and reduced product is Y , X and Y are respectively.

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Options:

A. SO_3, SO_2

B. SO_2, SO_2

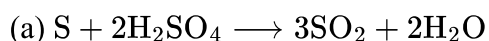
C. $\text{SO}_2, \text{H}_2\text{S}$

D. $\text{SO}_2, \text{H}_2\text{O}$

Answer: B

Solution:

The complete reaction is as follows,



Here, sulphur (S) is oxidised to SO_2

H_2SO_4 is reduced to SO_2 .

Hence, both X and Y are SO_2 .

Question13

Given below are two statements.

Statement I : In the decomposition of potassium chlorate Cl is reduced.

Statement II : Reaction of Na with O_2 to form Na_2O is a redox reaction.

The correct answer is

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Options:

A. Both statements I and II are correct.

B. Both statements I and II are not correct.

C. Statement I is correct but statement II is not correct.

D. Statement I is not correct but statement II is correct.

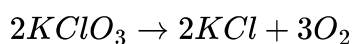
Answer: A

Solution:

Let's analyze each statement step by step:

For Statement I (Decomposition of potassium chlorate):

Consider the reaction:



In potassium chlorate ($KClO_3$), let's determine the oxidation state of chlorine.

For $KClO_3$, potassium (K) has an oxidation number of +1 and oxygen (O) is typically -2. Setting chlorine's oxidation number as x :

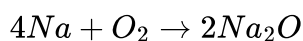
$$+1 + x + 3(-2) = 0 \Rightarrow x = +5$$

In potassium chloride (KCl), K is +1 and Cl must be -1 (to balance the compound).

Since chlorine goes from +5 to -1, it has gained electrons. This is a reduction. So Statement I is correct.

For Statement II (Reaction of sodium with oxygen to form Na_2O):

Consider the reaction:



Sodium (Na) starts with an oxidation state of 0 and in Na_2O , sodium is +1.

Oxygen (O_2) starts with 0 and in Na_2O , it is -2.

With sodium being oxidized (gain in oxidation number) and oxygen being reduced (loss in oxidation number), this clearly is a redox reaction. So Statement II is correct.

Since both statements are correct, the correct option is:

Option A

Both statements I and II are correct.

Question14

In neutral medium potassium permanganate oxidises I^- to X .. Identify the X .

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Options:

A. Iodine

B. Iodate

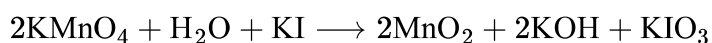
C. Per iodate

D. Hypo iodite

Answer: B

Solution:

In a neutral medium, potassium permanganate oxidizes I^- to IO_3^- (iodate). The balanced chemical reaction for this process is as follows:



In this reaction, the permanganate ion acts as an oxidizing agent, facilitating the conversion of iodide ions into iodate ions.

Question15

What are the oxidation numbers of S atoms in $S_4O_6^{2-}$?

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Options:

A. 6, -1, -1, 6

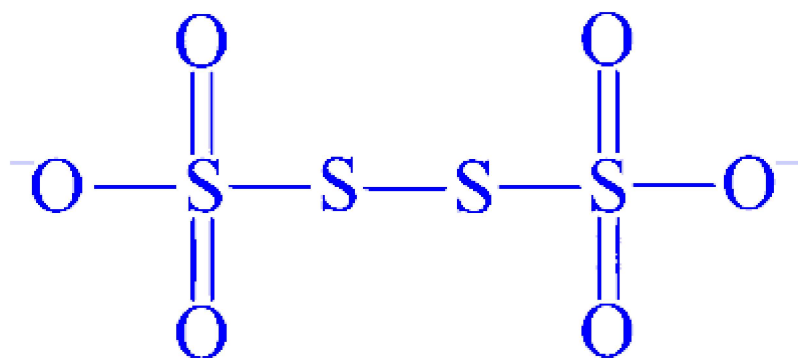
B. 5, 0, 0, 5

C. 2.5, 2.5, 2.5, 2.5

D. 7, -2, -2, 7

Answer: B

Solution:



When an atom is bonded to similar atoms, has zero oxidation state.

∴ Two middle sulphur atoms have zero oxidation state.

For other two sulphur atoms.

$$2x + 6 \times (-2) = -2$$

$$2x - 12 = -2$$

$$2x = -2 + 12$$

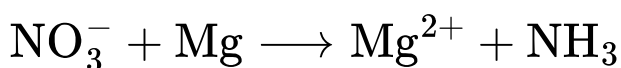
$$2x = +10$$

$$x = +5$$

∴ The oxidation number on S-atoms in $\text{S}_4\text{O}_6^{2-}$ is +5, 0, 0 + 5.

Question16

How many grams of Mg is required to completely reduce 100 mL, 0.1 M NO_3^- solution using the following reaction?



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Options:

A. 0.96

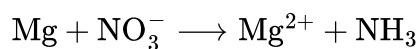
B. 0.62

C. 0.24

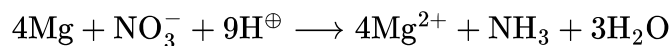
D. 0.75

Answer: A

Solution:



The complete balanced equation of the given reaction is



Given, conc. of $\text{NO}_3^- = 0.1 \text{ M}$

Volume of $\text{NO}_3^- = 100 \text{ mL}$

We know that,

$$\text{Molarity} = \frac{\text{Number of moles}}{\text{Volume of solution (in L)}}$$

Hence, number of moles = molarity $\times$ volume of solution (in L)

$$\begin{aligned}\text{So, number of moles} &= 0.1 \times 100 \times 10^{-3} \\ &= 10 \text{ m mol} \\ &= 10^{-2} \text{ mol}\end{aligned}$$

From above Eq., 1 mole NO_3^- reacts with 4 moles Mg. So, 10^{-2} mole NO_3^- reacts with 4×10^{-2} moles Mg. Now, to calculate the mass of Mg required, use the formula,

$$\text{Number of moles} = \frac{\text{Given mass}}{\text{Molar mass}}$$

$$\text{Molar mass of Mg} = 24 \text{ g mol}^{-1}$$

On substituting the value,

$$4 \times 10^{-2} = \frac{\text{Mass of Mg}}{24}$$

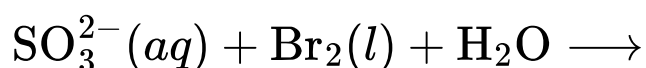
We will get,

$$\text{Mass of Mg} = 24 \times 4 \times 10^{-2}$$

$$\text{Mass of Mg} = 0.96 \text{ g}$$

Question17

What is the oxidation state of S in the sulphur containing product of the following reaction?



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Options:

A. +6

B. +4

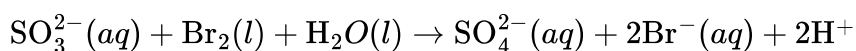
C. +2.5

D. +2

Answer: A

Solution:

The given reaction is:



To determine the oxidation state of sulfur (S) in the product, we first identify that the sulfur-containing product is SO_4^{2-} (sulfate ion). Let's denote the oxidation state of sulfur as x .

For the sulfate ion SO_4^{2-} :

- Each oxygen (O) atom has an oxidation state of -2 .
- There are four oxygen atoms, contributing a total of $4 \times (-2) = -8$.

The net charge on the sulfate ion is -2 . Using these data, we set up the following equation to solve for x :

$$x + 4(-2) = -2$$

Simplifying this:

$$x - 8 = -2$$

Solving for x :

$$x = -2 + 8$$

$$x = +6$$

Therefore, the oxidation state of sulfur in the sulfate ion SO_4^{2-} is $+6$.

Question18

In the reaction of phosphorus with conc. HNO_3 , the oxidised and reduced products respectively are

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Options:

A. $\text{H}_3\text{PO}_4, \text{NO}_2$

B. $\text{H}_3\text{PO}_2, \text{NO}$

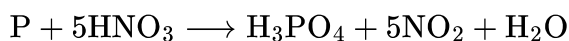
C. $\text{H}_3\text{PO}_3, \text{N}_2\text{O}$

D. HPO_3, NO

Answer: A

Solution:

Phosphorus reacts with HNO_3 to give H_3PO_4 and NO_2 . In this reaction, phosphorus is oxidised to H_3PO_4 and also N is reduced to NO_2 . The reaction is



Question19

What are the oxidation states of three Br atoms in Br_3O_8 molecule?

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Options:

A. +5, +6, +5

B. +6, +4, +6

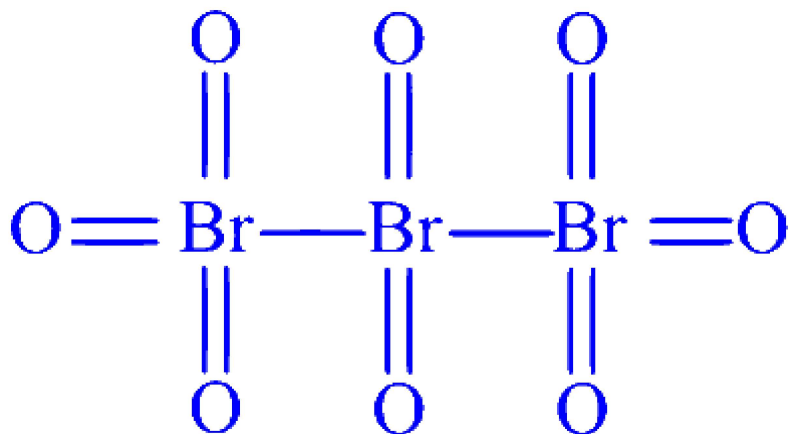
C. +7, +2, +7

D. +6 + 3, +7

Answer: B

Solution:

Structure of Br_3O_8 is



Each terminal Br has three oxygen atoms attached to it. So, the oxidation state of Br -atom on terminals is +6 .

The bromine atom in the middle has two oxygen atoms attached to it. So, the oxidation state of Br atom in middle is +4.

Hence, the oxidation states of three Br -atoms in Br_3O_8 molecule are +6, +4, +6.

Question20

Which of the following statement is incorrect?

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Options:

- A. Tl^{3+} salts are oxidising agents.
- B. Ga^{+} salts are reducing agents.
- C. Pb^{4+} salts are better oxidising agents.
- D. As^{+5} salts are better oxidising agents.

Answer: D

Solution:

(a) Tl is more stable in +1 oxidation state, then +3 oxidation state, therefore Tl^{3+} salts are oxidising agents.

(b) $\text{Ga}^{+} - 2e^{-} \rightarrow \text{Ga}^{3+}$ (More stable)

$\therefore \text{Ga}^+$ is a reducing agent.

(c) $\text{Pb}^{4+} + 2e^- \rightarrow \text{Pb}^{2+}$ (More stable)

Pb^{4+} is an oxidising agent.

(d) $\text{As}^{5+} + 2e^- \rightarrow \text{As}^{3+}$ (Less stable)

Thus, As^{5+} cannot be reduced so As^{5+} salts are not a good oxidising agent. An oxidising agent is defined as substance whose oxidation number decreases while reducing agent defined as substance whose oxidation number increases.

Question 21

Which among the following species does not show disproportionation reaction?

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Options:

A. ClO^-

B. ClO_2^-

C. ClO_3^-

D. ClO_4^-

Answer: D

Solution:

Disproportionation is a specific type of redox reaction in which a species is simultaneously reduced and oxidised to form two different products.

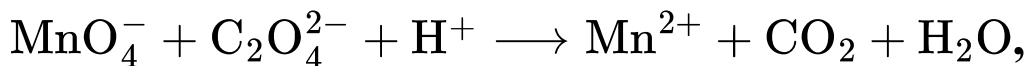
Electronic configuration of Cl is $1s^2 2s^2 2p^2 3s^2 3p^6 4s^1$. It can exhibit maximum oxidation state of +7.

All species can exhibit disproportionation reaction which simultaneously oxidise as well as reduce.

In ClO_4^- , Cl cannot undergo oxidation as +7 is maximum oxidation state exhibited by chlorine.

Question22

For the redox reaction



the correct coefficients of the reactants for the balanced reaction are respectively

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Options:

A. 2, 5, 16

B. 16, 3, 12

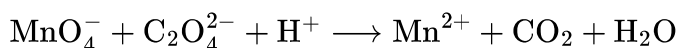
C. 15, 16, 12

D. 2, 16, 5

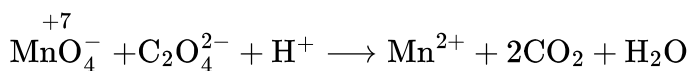
Answer: A

Solution:

The unbalanced redox reaction is

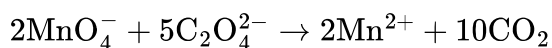


(i) Balanced C atoms



Oxidation state of Mn decreases from +7 to +2 .

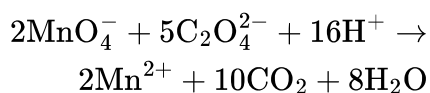
To balance the increase or decrease in oxidation number, multiply Mn containing species with 2 and C containing species with 5 .



(ii) Balance 'O' atoms by adding 8 water molecules to the product side.



(iii) Balance 'H' atoms by adding 16H^+ ions to the reactant side.



Hence, $\text{MnO}_4^- \Rightarrow 2$, $\text{C}_2\text{O}_4^{2-} \Rightarrow 5$; $\text{H}^+ = 16$

Question 23

Assertion (A) The colour of old lead paintings can be restored by washing them with a dilute solution of H_2O_2 .

Reason (R) Hydrogen peroxide reduces PbS to Pb .

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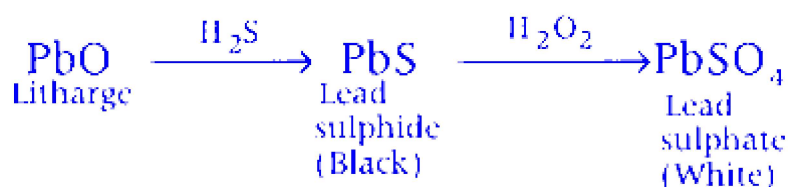
Options:

- A. Both A and R are true and R is a correct explanation for A .
- B. Both A and R are true but R is not a correct explanation for A .
- C. A is true but R is false.
- D. A is false but R is true.

Answer: C

Solution:

When the old oil painting are kept in the atmosphere for a longer time the tracer of gas present in atmosphere converts lead oxide (PbO) into PbS which is black in colour.



Therefore, these paintings get transhed. It can be restored by keeping then in H_2O_2 solution for sometime as a result of which lead sulphide is oxidised to lead sulphate.

So, A is true but R is false.

Question24

Which among the following species acts as a self-indicator?

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Options:

A. H_2O_2

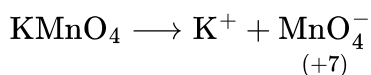
B. I^-

C. $\text{Cr}_2\text{O}_7^{2-}$

D. MnO_4^-

Answer: D

Solution:



In presence of a reducing agent, MnO_4^- acts as a self-indicator its end point is pink to colourless. This is because in MnO_4^- , Mn is in +7 oxidation state and is in highest oxidation state. Thus, it tends to get reduced and easily takes electrons and being a charge transfer (CT) complex, shows intense colour.

After titration, it is MnO_2 in +4 state and has no electronic charge transfer, therefore colourless.
